

The Rank-Defect Theorem

The generalization beyond the ring: which stationary processes satisfy the equivalence of the arrow of time with a Hankel rank defect. Soundness holds — a visible complex mode is a sound arrow witness — but off the abelian slice both failure modes are generic: hidden arrows on real-spectrum chains (positive measure in the Birkhoff polytope) and fake arrows in the Gaussian stratum. On the N-ring the reversible slice is exactly the spectral-merging locus, the rank doubles with the arrow, and the reflection-laundering theorem shows the ring's arrow is its chirality: the **Amin Abae** the arrow exactly when it loses the handedness. With the chat's Experiments 1–3 run at full audit strength: the AAK interval exact, the register explosion $9 \rightarrow 9153$, the sheaf ordering perfect at the task level and null at the seed level.